did happen was that, as in the artist's impression above, they were able to cross a gap in a ridge by holding on to each other. They aggregated; they moved in a co-ordinated fashion; they lifted heavy objects, as promised; they navigated rough terrain; and did many other things worthy of a second round of applause. If you're not on dial-up, see www.swarm-bots.org/index.php?main=3&sub=35&conpage=s41r for videos of s-bot(s) on various kinds of terrain.

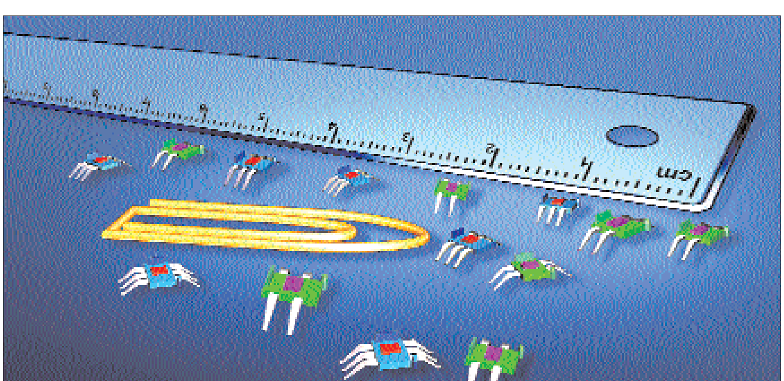
The Real Deal

If a swarm is to do undersea exploration, you can't have a piddling ten s-bots holding hands and singing. What you need is a *real* swarm, with the measurements in the nanometres and the numbers in the millions—read nanotech.

Intelligent Small World Autonomous Robots for Micro-manipulation, or I-Swarm, is a project at the Institute for Process Control and Robotics of the Universität Karlsruhe in Germany. It aims at facilitating the mass-production of microbots. These are expected to be deployed as swarms consisting of up to a thousand units, and the researchers expect the swarm to be able to do lots of things, including micro-assembly and biological, medical, and cleaning tasks. (No, these bots won't be sent to Mars either.) Amongst the broad objectives of the project are to demonstrate collective task execution, to showcase the collective intelligence of robots, and to "use advanced sensors and tools for manipulation in the small world."

The artist's impression below is of the bots being designed at I-Swarm. They're a few millimetres large, and are pretty basic in structure as well as capabilities. This project extends to 2008: be sure to check back in this space to find out what happened! If your band is broad enough, visit http://dsc.discovery.com/news/media/swarmrobot-video.html for a video of cute little swarmbots doing equally cute things.

Then, there's the Smart Dust project on at the University of California at Berkeley. This one aims at building a "millimetre-scale sensing and communication platform for a massively distributed sensor network. This device will be around the size of a grain of sand and will contain sensors and bi-directional wireless communications, while being inexpensive enough to deploy by the hundreds." So what would the grain-sized sensing device do when deployed by the hundreds? Lots of things, possibly.



The bots being designed at the I-Swarm project look very much like adorable insects

"Yes, personal privacy is getting harder and harder to come by. Yes, you can hype Smart Dust as being great for big brother. Yawn. Every technology has a dark-side deal with it

Kris Pister
Professor of Electrical Engineering and Computer Science
University of California at Berkeley

Of Devils And Dust

Kris Pister, Professor of Electrical Engineering and Computer Science at the University of California at Berkeley, has imagined several applications for Smart Dust, his baby. Of course, each of us is free to imagine, and if you've got what seems a good idea, write in to Pister—he's listening!

Think of this: you could glue a Smart Dust mote to each of your fingernails. Then, accelerometers will sense the orientation and motion of your fingertips, and talk to your computer. If it knows where your fingers are, you could sculpt 3D shapes in virtual clay, play the piano, and do much more!

Then, there can be defence-related sensor networks that can perform battlefield surveillance, and transportation monitoring. Smart Dust could find its way into quality monitoring: things such as temperature, humidity of meat and dairy products, and so on, could be monitored.

Inventory control applications are interesting: "The carton talks to the box, the box talks to the palette, the palette talks to the truck, and the truck talks to the warehouse, and the truck and the warehouse talk to the Internet. Know where your products are and what shape they're in any time, anywhere."

You could put Smart Dust motes on a quadriplegic's face, "to monitor blinking and facial twitches—and send them as commands to a wheelchair / computer / other device." (This was an idea someone sent in.)

Pister also speaks of smart office spaces, where ambient conditions are tailored to the needs of each individual. "Maybe soon we'll all be wearing temperature, humidity, and environmental comfort sensors sewn into our clothes, continuously talking to our workspaces which will deliver conditions tailored to our needs."

Naturally, many others, including us, have tried to imagine what use smart dust, or a swarm of motes, or a roboswarm, or whatever, could be put to. If a swarm were to explore the ocean floor, it could be programmed to look for precious metals or minerals. When one unit finds something that seems precious, it could send out signals that would cause others to gravitate to that spot—but not all of them, because the area under surveillance has to be populated. Then, if a few bots get eaten by a shark, their absence would be felt, and others would come in to fill the gap.

Think of traffic surveillance: if you needed to know the general direction of traffic, or the traffic pattern in a particular area, you could just spray the entire area with a near-invisible swarm of cam-equipped bots! You'd get the entire picture—including, if you wanted such data, where there are more yellow cars and where the humidity correlates with accidents—without the need for complex computer vision algorithms, and with the entire swarm reprogrammable at the press of a button.

Think of medical applications. Nanobots could do things like repairing clogged arteries, and a variety of procedures that involve going where medical equipment can't go directly. Think of dangerous missions like clearing out minefields. Think of operations that involve going where no man has gone before. What seems a better idea—sending in a programmed humanoid and risking